XLVII National Mathematics Olympiad: 3rd round, 25–26 April 1998

Problem 1. Find the least positive integer number n $(n \geq 3)$ with the following property: for any colouring of n different points $A_1, A_2, \ldots, A_n$ on a line and such that $A_1A_2 = A_2A_3 = \ldots = A_{n-1}A_n$ in two colours, there are three points $A_i, A_j, A_{2j-i} (1 \leq i < 2j - i \leq n)$ which have the same colour.

Solution: Assume the two colours are white and black. Consider 8 points coloured as follows: A_1, A_2, A_5, A_6 (white), A_3, A_4, A_7, A_8 (black). Obviously no three points A_i, A_j, A_{2j-i} $(1 \leq i < 2j - i \leq n)$ have the same colour and therefore $n \geq 9$.

If we can show that $n = 9$ has the required property, we will be done. Suppose there are 9 points coloured black or white and no three points A_i, A_j, A_{2j-i} $(1 \leq i < 2j - i \leq n)$ have the same colour.

First assume that for $i = 3$, $i = 4$ or $i = 5$ points A_i and A_{i+2} have the same colour (say white). Then the points $A_{i-2}, A_{i+1}, A_{i+4}$ should be black (note that $i - 2 \geq 1$ and $i + 4 \leq 9$), which is a contradiction.

Suppose now that for $i = 3, 4, 5$ the points A_i and A_{i+2} have different colours. Without loss of generality assume A_5 is a white point. Then A_3 and A_7 are black. Because of the symmetry we may suppose that A_4 is white and A_6 is black. Consiquently A_8 is white, A_2 is black $(2 + 8 = 2 \cdot 5)$ and A_9 is white $(7 + 9 = 2 \cdot 8)$. Therefore A_1 should be both white $(1 + 3 = 2 \cdot 2)$ and black $(1 + 9 = 2 \cdot 5)$, which is again a contradiction.

Consequently the assumption is not true and so $n = 9$.

Problem 2. Let $ABCD$ be a quadrilateral such that $AD = CD$ and $\angle DAB = \angle ABC < 90°$. The line passing through D and the midpoint of the segment BC intersects the line AB at the point E. Prove that $\angle BEC = \angle DAC$.

Solution: Let M be the midpoint of BC and let AD and BC meet at point N and AN and EC meet at point P. It follows from Menelaus' Theorem aplied to $\triangle DMN$ and $\triangle DEN$ that $DP \cdot NC \cdot ME = PN \cdot CM \cdot ED$ and $DA \cdot NB \cdot ME = AN \cdot BM \cdot ED$. Combining the above equalities with $AN = BN$, $BE = CE$ and $AD = CD$, we get $DP \cdot NC = DC \cdot PN$. Therefore CP is bisector of $\angle DCN$.

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Consequently $\angle ACP = \angle ACD + \angle DCP = \frac{1}{2}(\angle NDC + \angle DCN) = \angle NAB$ so $\angle DCP = \angle CAB$ and we obtain $\angle BEC = \angle ABC - \angle BCE = \angle BAD - \angle DCP = \angle DAC$, *Q. E. D.*

Note: The assertion is also true if the condition $\angle ABC < 90°$ is left out.

Problem 3. Let $\mathbb{R}^+$ be the set of all positive real numbers. Prove that there is no function $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$(f(x))^2 \geq f(x+y)(f(x)+y)$$

for arbitrary positive real numbers x and y.

Solution: Suppose there exists a function satisfying the conditions of the problem. Write the initial equality in the form

$$f(x) - f(x+y) \geq \frac{f(x)y}{f(x)+y}.$$

First we prove that $f(x) - f(x+1) \geq \frac{1}{2}$ for $x > 0$. Obviously f is a (strictly) monotone non-increasing function. Fix $x > 0$ and choose a natural number n, such that $n \cdot F(x+1) \geq 1$. When $k = 0, 1, \ldots, n-1$, we obtain that

$$f(x + \frac{k}{n}) - f(x + \frac{k+1}{n}) \geq \frac{f(x+\frac{k}{n})\frac{1}{n}}{f(x+\frac{k}{n}) + \frac{1}{n}} \geq \frac{1}{2n}.$$

Adding the above inequalities gives $f(x) - f(x+1) \geq \frac{1}{2}$.

Let the natural number m be such that $m \geq 2f(x)$. Therefore

$$f(x) - f(x+m) = \sum_{i=0}^{m-1} (f(x+i) - f(x+i+1)) \geq \frac{m}{2} \geq f(x),$$

and so $f(x+m) \leq 0$. But this contradicts the fact that f is strictly positive.

Problem 4. Let $f(x) = x^3 - 3x + 1$. Find the number of different real solutions of the equation $f(f(x)) = 0$.

Solution: Since $f'(x) = 3(x-1)(x+1)$ it follows that f is strictly monotone non-decreasing in the intervals $(-\infty, -1]$ and $[1, \infty)$ and strictly monotone non-increasing in the interval $[-1, 1]$. Moreover $\lim\limits_{x \to \pm\infty} f(x) = \pm\infty, f(-1) = 3, f(1) = -1, f(3) = 19 > 0$ and so the equation $f(x) = 0$ has three distinct roots x_1, x_2, x_3 such that $x_1 < -1 < x_2 < 1 < x_3 < 3$. Therefore $f(x) = x_1$ has only one real root (which is less than -1) and $f(x) = x_2$ and $f(x) = x_3$ have three distinct real roots each (one in each of the intervals $(-\infty, -1), (-1, 1)$ and $(1, \infty)$). Since the roots of $f(f(x)) = 0$ are exactly the roots of these three equations, we conclude that it has seven distinct real roots.

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Problem 5. The convex pentagon $ABCDE$ is inscribed in a circle with radius R. The inradii of the triangles ABC, ABD, AEC and AED are denoted by r_{ABC}, r_{ABD}, r_{AEC} and r_{AED}. Prove that

a.) $\cos \angle CAB + \cos \angle ABC + \cos \angle BCA = 1 + \dfrac{r_{ABC}}{R}$;

b.) If $r_{ABC} = r_{AED}$ and $r_{ABD} = r_{AEC}$, then $\triangle ABC \cong \triangle AED$.

Solution: a.) Using the standard notation for the elements of $\triangle ABC$, we obtain that

$$\frac{r}{R} = \frac{S}{pR} = \frac{4S^2}{pabc} = \frac{4(p-a)(p-b)(p-c)}{abc} =$$

$$= \frac{1}{2abc}(a(b^2+c^2-a^2) + b(c^2+a^2-b^2) + c(a^2+b^2-c^2) - 2abc) =$$

$$= \cos \angle CAB + \cos \angle ABC + \cos \angle BCA - 1.$$

b.) Applying continuously the equality from a.) and using the fact that $ABCDE$ is inscribed in a circle convex pentagon, it is easy to see that

$$r_{ABC} + r_{AEC} + r_{EDC} = r_{AED} + r_{ABD} + r_{BCD}.$$

The condition of the problem and the above equality imply $r_{BCD} = r_{EDC}$. Since $\triangle BCD$ and $\triangle EDC$ have a side in common and equal circumradiuses we get that $\triangle BCD \cong \triangle EDC$ (prove it using a.)). In particular $BC = ED$ and using $r_{ABC} = r_{AED}$ again, we get $\triangle ABC \cong \triangle AED$.

Problem 6. Show that the equation

$$x^2y^2 = z^2(z^2 - x^2 - y^2)$$

has no solution in positive integer numbers.

Solution: Suppose x, y, z is a solution for which $\dfrac{xy}{z}$ (which is a natural number—why?) has its minimum value.

Write x, y, z in the form $x = dx_1, y = dy_1, z = dz_1$, where $d = (x, y, z)$. Our equation is equivalent to $x_1^2y_1^2 = z_1^2(z_1^2 - x_1^2 - y_1^2)$. Let $u = (x_1, z_1), v = (y_1, z_1), x_1 = ut, y_1 = vw$. Since z_1 divides x_1y_1, we get $z_1 = uv$. A substitution in the last equality gives

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Without loss of generality we may assume that the first pair of equalities hold. It is easily seen that v and u are odd integer numbers. It follows now from $u^2 + w^2 = v^2$ that $u = m^2 - n^2, w = 2mn, v = m^2 + n^2$, where m and n are coprime natural numbers (of distinct parity). Substitution in $v^2 + t^2 = 2u^2$ shows that $t^2 + (2mn)^2 = (m^2 - n^2)^2$ and so $t = p^2 - q^2, mn = pq, m^2 - n^2 = p^2 + q^2$ for some natural numbers p and q. Therefore

$$p^2q^2 = m^2(m^2 - p^2 - q^2),$$

which shows that p, q, m is a solution of the original equation. It remains to be seen that $\frac{pq}{m} = n < d(p^2 - q^2)2mn = \frac{xy}{z}$, which contradicts the way we have chosen x, y, z.

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